

-- Module ACSE-1 (X.227:04/1995)
 -- See also [ITU-T X.227 \(04/1995\)](#)
 -- See also the [index](#) of all ASN.1 assignments needed in this document

ACSE-1 {joint-iso-itu-t association-control(2) modules(0) apdus(0) version1(1)}

-- *ACSE-1 refers to ACSE version 1*

DEFINITIONS ::=

BEGIN

EXPORTS

acse-as-id, ACSE-apdu, aCSE-id, Application-context-name, AP-title,
 AE-qualifier, AE-title, AP-invocation-identifier, AE-invocation-identifier,
 Mechanism-name, Authentication-value, ACSE-requirements, ObjectSet;

IMPORTS

Name, RelativeDistinguishedName
FROM InformationFramework {joint-iso-itu-t ds(5) module(1)
 informationFramework(1) 3};

-- *The data types Name and RelativeDistinguishedName are imported from ISO/IEC 9594-2.*

-- *object identifier assignments*

acse-as-id **OBJECT IDENTIFIER** ::=
 {joint-iso-itu-t association-control(2) abstract-syntax(1) apdus(0)
 version1(1)}

-- *may be used to reference the abstract syntax of the ACSE APDUs*

aCSE-id **OBJECT IDENTIFIER** ::=
 {joint-iso-itu-t association-control(2) ase-id(3) acse-ase(1) version(1)}

-- *may be used to identify the Association Control ASE.*

-- *top level CHOICE*

ACSE-apdu ::= **CHOICE** {

aarq [AARQ-apdu](#),
 aare [AARE-apdu](#),
 rlrq [RLRQ-apdu](#),
 rlre [RLRE-apdu](#),
 abrt [ABRT-apdu](#),

...

}

AARQ-apdu ::= [APPLICATION 0] **IMPLICIT SEQUENCE** {
 protocol-version
 [0] **IMPLICIT BIT STRING** {version1(0)} **DEFAULT** {version1},
 application-context-name [1] [Application-context-name](#),
 called-AP-title [2] [AP-title](#) **OPTIONAL**,

```

called-AE-qualifier           [3] AE-qualifier OPTIONAL,
called-AP-invocation-identifier [4] AP-invocation-identifier OPTIONAL,
called-AE-invocation-identifier [5] AE-invocation-identifier OPTIONAL,
calling-AP-title               [6] AP-title OPTIONAL,
calling-AE-qualifier           [7] AE-qualifier OPTIONAL,
calling-AP-invocation-identifier [8] AP-invocation-identifier OPTIONAL,
calling-AE-invocation-identifier [9] AE-invocation-identifier OPTIONAL,
-- The following field shall not be present if only the Kernel is used.
sender-acse-requirements       [10] IMPLICIT ACSE-requirements OPTIONAL,
-- The following field shall only be present if the Authentication functional unit is selected.
mechanism-name                 [11] IMPLICIT Mechanism-name OPTIONAL,
-- The following field shall only be present if the Authentication functional unit is selected.
calling-authentication-value    [12] EXPLICIT Authentication-value OPTIONAL,
application-context-name-list   [13] IMPLICIT Application-context-name-list OPTIONAL,
-- The above field shall only be present if the Application Context Negotiation functional unit is selected
implementation-information      [29] IMPLICIT Implementation-data OPTIONAL,
...,
...,
user-information
  [30] IMPLICIT Association-information OPTIONAL
}

```

```

AARE-apdu ::= [APPLICATION 1] IMPLICIT SEQUENCE {
  protocol-version
    [0] IMPLICIT BIT STRING {version1(0)} DEFAULT {version1},
  application-context-name
    [1] Application-context-name,
  result
    [2] Associate-result,
  result-source-diagnostic
    [3] Associate-source-diagnostic,
  responding-AP-title
    [4] AP-title OPTIONAL,
  responding-AE-qualifier
    [5] AE-qualifier OPTIONAL,
  responding-AP-invocation-identifier [6] AP-invocation-identifier OPTIONAL,
  responding-AE-invocation-identifier [7] AE-invocation-identifier OPTIONAL,
-- The following field shall not be present if only the Kernel is used.
responder-acse-requirements       [8] IMPLICIT ACSE-requirements OPTIONAL,
-- The following field shall only be present if the Authentication functional unit is selected.
mechanism-name
    [9] IMPLICIT Mechanism-name OPTIONAL,
-- This following field shall only be present if the Authentication functional unit is selected.
responding-authentication-value
    [10] EXPLICIT Authentication-value OPTIONAL,
application-context-name-list
    [11] IMPLICIT Application-context-name-list OPTIONAL,
-- The above field shall only be present if the Application Context Negotiation functional unit is selected
implementation-information
    [29] IMPLICIT Implementation-data OPTIONAL,
...,
...,
}

```

```

    user-information
    [30] IMPLICIT Association-information OPTIONAL
}

RLRQ-apdu ::= [APPLICATION 2] IMPLICIT SEQUENCE {
    reason          [0] IMPLICIT Release-request-reason OPTIONAL,
    ...,
    ...,
    user-information [30] IMPLICIT Association-information OPTIONAL
}

RLRE-apdu ::= [APPLICATION 3] IMPLICIT SEQUENCE {
    reason          [0] IMPLICIT Release-response-reason OPTIONAL,
    ...,
    ...,
    user-information [30] IMPLICIT Association-information OPTIONAL
}

ABRT-apdu ::= [APPLICATION 4] IMPLICIT SEQUENCE {
    abort-source      [0] IMPLICIT ABRT-source,
    abort-diagnostic  [1] IMPLICIT ABRT-diagnostic OPTIONAL,
    -- This field shall not be present if only the Kernel is used.
    ...,
    ...,
    user-information [30] IMPLICIT Association-information OPTIONAL
}

ABRT-diagnostic ::= ENUMERATED {
    no-reason-given(1), protocol-error(2),
    authentication-mechanism-name-not-recognized(3),
    authentication-mechanism-name-required(4), authentication-failure(5),
    authentication-required(6), ...
}

ABRT-source ::= INTEGER {acse-service-user(0), acse-service-provider(1)}
(0..1, ...)

ACSE-requirements ::= BIT STRING {
    authentication(0), application-context-negotiation(1)}

Application-context-name-list ::= SEQUENCE OF Application-context-name

Application-context-name ::= OBJECT IDENTIFIER

-- Application-entity title productions follow (not in alphabetical order)
AP-title ::= CHOICE {
    ap-title-form1 AP-title-form1,

```

```

    ap-title-form2 AP-title-form2,
    ...
}

```

```

AE-qualifier ::= CHOICE {
    ae-qualifier-form1 AE-qualifier-form1,
    ae-qualifier-form2 AE-qualifier-form2,
    ...
}

```

-- When both AP-title and AE-qualifier data values are present in an AARQ or AARE APDU, both must have the same form to allow the construction of an AE-title as discussed in CCITT Rec. X.665 | ISO/IEC 9834-6.

```

AP-title-form1 ::=

```

[Name](#)

-- The value assigned to AP-title-form1 is The Directory Name of an application-process title.

```

AE-qualifier-form1 ::=

```

[RelativeDistinguishedName](#)

-- The value assigned to AE-qualifier-form1 is the relative distinguished name of a particular application-entity of the application-process identified by AP-title-form1.

```

AP-title-form2 ::= OBJECT IDENTIFIER

```

```

AE-qualifier-form2 ::= INTEGER

```

```

AE-title ::= CHOICE {
    ae-title-form1 AE-title-form1,
    ae-title-form2 AE-title-form2,
    ...
}

```

-- As defined in CCITT Rec. X.650 | ISO 7498-3, an application-entity title is composed of an application-process title and an application-entity qualifier. The ACSE protocol provides for the transfer of an application-entity title value by the transfer of its component values. However, the following data type is provided for International Standards that reference a single syntactic structure for AE titles.

```

AE-title-form1 ::=

```

[Name](#)

-- For access to The Directory (ITU-T Rec. X.500-Series | ISO/IEC 9594), an AE title has AE-title-form1. This value can be constructed from AP-title-form1 and AE-qualifier-form1 values contained in an AARQ or AARE APDU. A discussion of forming an AE-title-form1 from AP-title-form1 and AE-qualifier-form1 may be found in CCITT Rec. X.665 | ISO/IEC 9834-6.

```

AE-title-form2 ::= OBJECT IDENTIFIER

```

-- A discussion of forming an AE-title-form2 from AP-title-form2 and AE-qualifier-form2 may be found in CCITT Rec. X.665 | ISO/IEC 9834-6.

```
-- The abstract syntax of (calling/responding) authentication-value is determined by the authentication
-- mechanism used during association establishment. The authentication mechanism is either explicitly
-- denoted by the &id field (of type OBJECT IDENTIFIER) for a mechanism belonging to the class
-- MECHANISM-NAME, or it is known implicitly by
```

-- *prior agreement between the communicating partners. If the "other" component is chosen, then*
-- *the "mechanism-name" component must be present in accordance with*
-- *ITU-T Rec. X.680 | ISO/IEC 8824. If the value "mechanism-name" occurs in the AARQ-apdu or the*
-- *AARE-apdu, then that value must be the same as the value for "other-mechanism-name"*

Implementation-data ::= **GraphicString**

Mechanism-name ::= **OBJECT IDENTIFIER**

MECHANISM-NAME ::= **TYPE-IDENTIFIER**

ObjectSet MECHANISM-NAME ::=
{...}

Release-request-reason ::= **INTEGER** {normal(0), urgent(1), user-defined(30)
}(0 | 1 | 30, ...)

Release-response-reason ::= **INTEGER** {
normal(0), not-finished(1), user-defined(30)}(0 | 1 | 30, ...)

END

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